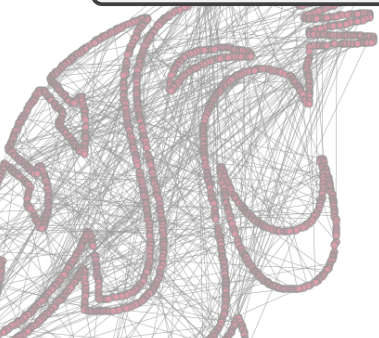


Graphgebrability & Mathematical Exploration

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April 20th, 2026



Outline

① Monte Carlo

- Accept/Reject

- Probability Theory

- Role in Gerrymandering

② Markov Chain Monte Carlo & Sampling of Clustered Graphs

- MCMC

- Linear Algebra

- Sampling Similarly Clustered Graphs

*A lot of this was a run through of code. You can find the Jupyter notebook and run through the code yourself by clicking [here](#): link

What We'll Discuss

We'll cover some of the following at a vague level. If you are interested, we can always discuss the nitty gritty details afterward!

- Probability – Law of Large Numbers and Markov Chains
- Graph Theory – Basics and Laplacian
- Linear Algebra – Eigenvectors and eigenvalues

What We'll See

We will see graphs denoted $G = (V, E)$. We will assume

- G is simple (no self-loops and no multiple edges)
- G is undirected (if a node i is connected to a node j , j is connected to i)
- G is unweighted (the existing edges all have the same weight, 1)

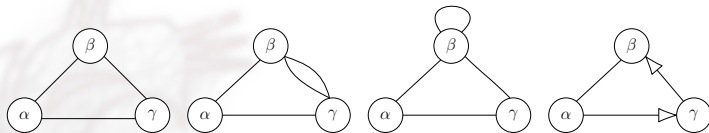


Figure: On left, a simple, undirected, unweighted graph. Everything else: bad.

Erdős-Renyi

As an introduction to random graphs, one fundamental random graph model is the Erdős-Renyi Model:

1. Each possible edge (i,j) has probability p of appearing in the graph.
2. We will disregard the resulting graph if disconnected and redraw.

Google Collab time!

① Monte Carlo

Accept/Reject

Probability Theory

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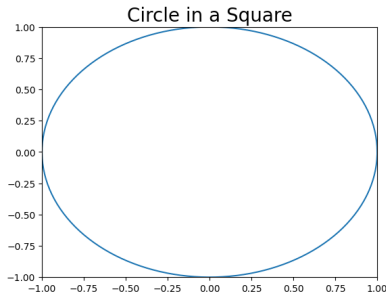
Accept/Reject

One exceptionally useful sampling strategy is the *Monte Carlo method* also called *Accept/Reject method*.

Monte Carlo Method

- Generate a random proposal sample
- Accept the sample if desirable and reject if not

A classic example is in estimating π using random points!



① Monte Carlo

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Questions

Question

- What is the *average* of $\{1, 2, 3, 4, 5\}$?
- What is the *expected value* of a dice roll with faces $\{1, 2, 3, 4, 5\}$?

Intuitively

3 for both!

Complicatedly

- The average is the sum $\frac{1}{5}(1 + 2 + 3 + 4 + 5) = 3$
- The *average of the values we expect to see* is 3.5
($\mathbb{E}[X] = \frac{1}{5}1 + \frac{1}{5}2 + \frac{1}{5}3 + \frac{1}{5}4 + \frac{1}{5}5 = 3$ where X is the outcome of a 5-sided dice roll)

Let's Roll Some Dice!

What we will do:

- Raise your hands (or shout out) with a number 1 through 5
- I will plot the averages
- I will compute the amount of times we see a specific value

<https://garrett-kepler.github.io/resources/LLN/>

Weak Law of Large Numbers (WLLN)

Intuitively

The average should be close to the expected value eventually. That is, if we roll a 5-sided dice a lot, we should see on average something close to 3 as we continue to roll.

Less Intuitively

The average $\frac{1}{n} \sum_{i=1}^n X_i$ will probably converge to $\mathbb{E}[X_i]$ as $n \rightarrow \infty$

Complicatedly

If $X_1, X_2, \dots$ are a sequence of independent identically distributed random variables with $\mathbb{E}[X] < \infty$, then for any $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X_i] \right| < \epsilon \right] = 1$$

Graph Reframing

This also applies to graphs as well!

For $n = 1, 2, 3, \dots$

- Take a random graph G_n
- Compute some statistic $f(G_n)$ on the graph and compute the average so far $\sum_{i=1}^n f(G_i)$
- Repeat

We know $\sum_{i=1}^n f(G_i)$ will converge to the expected value of $f(G_i)$ at some point!

[Link here to collab](#)

A New School of Thought



Figure: (Left) Pavel Nekrasov: firm believer WLLN only applies to independent events. (Right) Andrey Markov: prover that WLLN applies to dependent events too!

WLLN with Dependence

Intuitively

The beautiful result Markov proved is that we actually don't need independent events! With a little assumptions we can confidently say the average will get close to the expected value.

Complicatedly

If $X_1, X_2, \dots$ is a positive Harris recurrent Markov chain with unique invariant distribution π and f is a real-valued function on the state space, then

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{a.s.} \mu = \mathbb{E}_{\pi} f(X) = \int f(x) \pi(dx)$$

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Redistricting

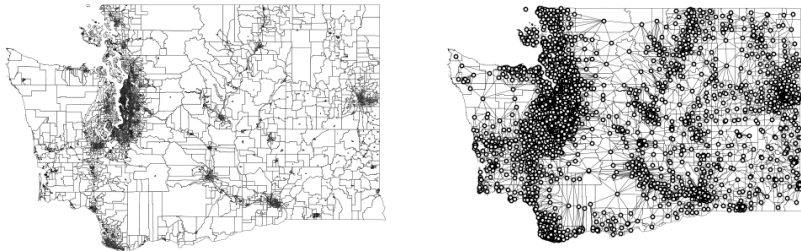


Figure: Census tract level cuts (dual graph on right)

Redistricting

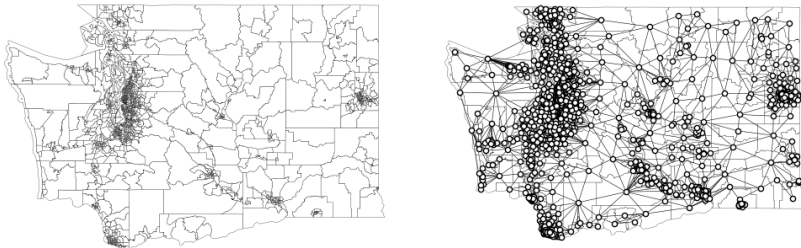


Figure: Census block level cuts (dual graph on right)

Redistricting



Figure: Legislative districts decided on in 2020 (dual graph on right)

Ensemble Method

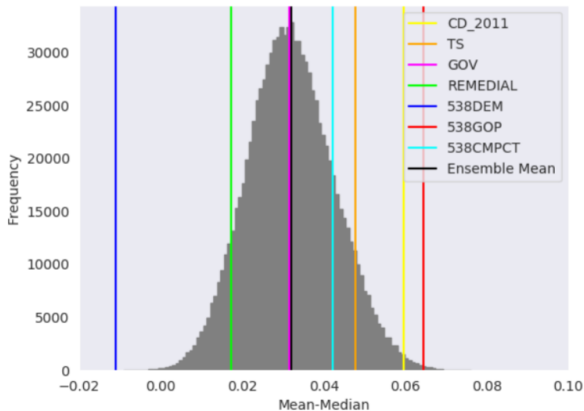


Figure: An example of ensemble analysis in action. Note, we have generated *many, many* samples in hopes the average is close to the expected value as the LLN guarantees.

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MCMC

Linear Algebra

Sampling Similarly Clustered Graphs

① Monte Carlo

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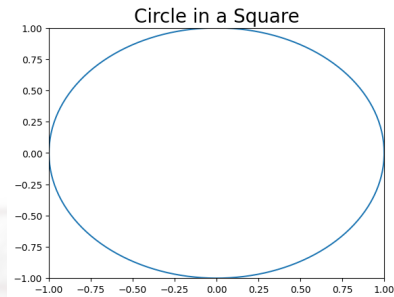
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MCMC on the Circle

Coming back to our circle inscribed in a square:



We generated independent and uniformly random points. What if we preferred points in a specific region?

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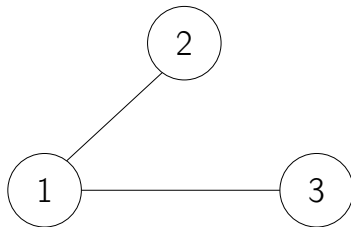
Let's Make a Matrix

Definition

The *Laplacian* of G is the $n \times n$ matrix $L = L(G)$ with entries

$$L_{ij} = \begin{cases} \# \text{ of edges attached to } i & \text{if } i = j \\ -1 & \text{if } i \text{ connected to } j \text{ and } i \neq j \\ 0 & \text{otherwise} \end{cases}$$

Example



$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Spectral Information

What we really care about are the eigenvectors and eigenvalues of L : $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ and $x_1, x_2, \dots, x_n$.

If the graph is connected,

- $\lambda_2 > 0$
- the corresponding eigenvector x_2 can be used to partition the graph into two connected components

Theorem (Fiedler Partitioning)

Let G be a connected graph and let x_2 be its corresponding Fiedler vector. Then, the subgraphs induced by the vertex sets $V_{\geq 0} = \{i : x(i) \geq 0\}$ and $V_- = V \setminus V_{\geq 0}$ are connected [?].

Fiedler Partitioning

The way it works:

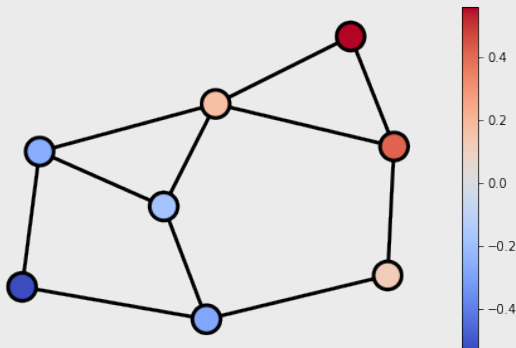
- Take the second eigenvector x_2 .
- Find the non-negative entries $V_{\geq 0} = \{i \mid x_2(i) \geq 0\}$ and the negative entries $V_{< 0} = \{i \mid x_2(i) < 0\}$
- Look at the subgraphs on $V_{\geq 0}$ and $V_{< 0}$
- They will be connected!

Fiedler Partitioning

Example

An instance of the Erdős-Renyi graph partitioned using its Fiedler vector

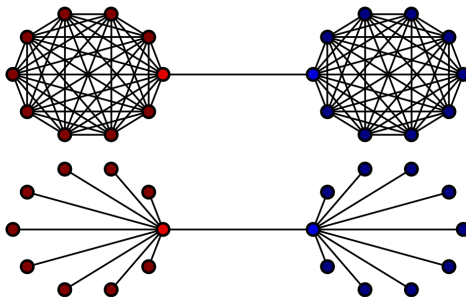
$$x_2 \approx [-0.29, 0.42, 0.56, -0.26, -0.53, -0.19, 0.17, 0.12]^T$$



Fiedler Partitioning

Example

$$x \approx [0.23, 0.23, \dots, 0.23, 0.19, -0.19, -0.23, -0.23, \dots, -0.23]$$
$$\lambda \approx 0.169$$

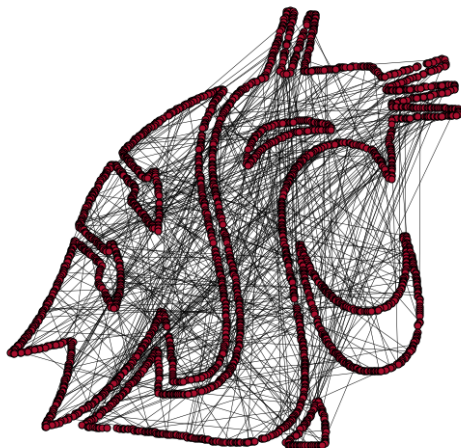


Bonus

Any toggling of edges within the clusters with shared assignment will preserve the eigenpair. Also works for weighted graphs!

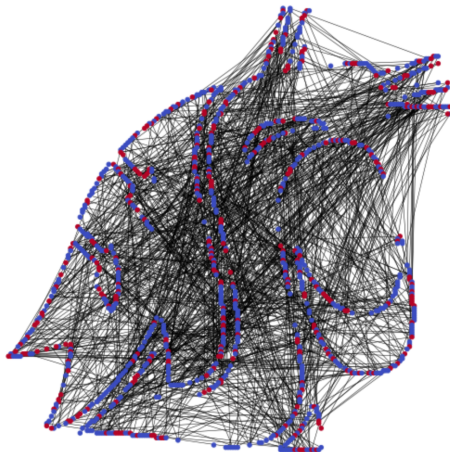
Fiedler Partitioning

Example



Fiedler Partitioning

Example



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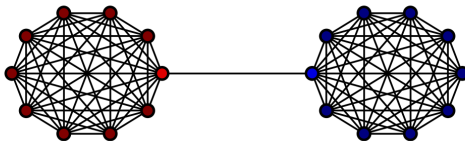
Problem

Question

Given the space of connected graphs $\mathcal{G}$, how can we randomly sample graphs that are “similarly clustered”?

Example

Given the following clustering on the barbell graph, how can we randomly sample graphs with this same clustering?



Graph Metropolis-Hastings

To generate samples from some desired distribution $f(\cdot)$, we can run the following:

M.H. Algorithm (Symmetric)

- Initialize: Let f be a function proportional to some target distribution π over the set of connected graphs $\mathcal{G}$. Start with some arbitrary graph $G_t \in \mathcal{G}$.
- Iterate: For each iteration do the following
 - Propose some new state G' by randomly toggling a subset of edges $H' \subseteq G_t$.
 - If G' is disconnected, reject. Otherwise:
 - Let $\alpha = \min \left\{ \frac{f(G')}{f(G_t)}, 1 \right\}$.
 - Generate $\beta \sim U(0, 1)$.
 - If $\beta \leq \alpha$, accept: $G_{t+1} = G'$. Otherwise, reject: $G_{t+1} = G_t$.

Unbiased Proposal Step

The cleverness comes in picking f . For example, picking f such that

$$f(G_t) = \begin{cases} \frac{1}{|\mathcal{G}|} & \text{if } G_t \text{ is connected} \\ 0 & \text{otherwise} \end{cases}$$

targets the uniform distribution over connected graphs.

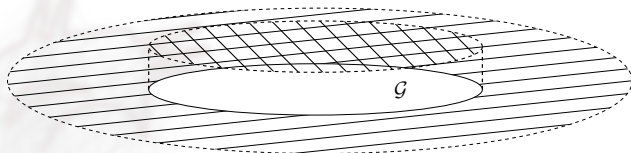


Figure: Target distribution over space of connected graphs $\mathcal{G}$

Uniform Target Distribution

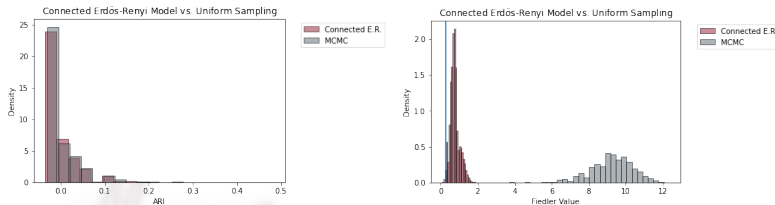


Figure: Results of 10 chains with 10,000 samples each targeting the uniform distribution. $\hat{R} \approx .81$. $ESS \approx 2,100$.

Scenario in Question

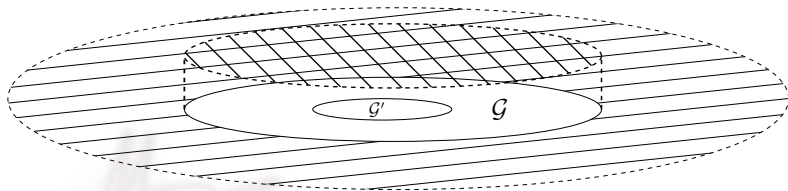


Figure: The uniform distribution over the set of connected graphs $\mathcal{G}$ with the subset of graphs we want, $\mathcal{G}'$

Target Distribution

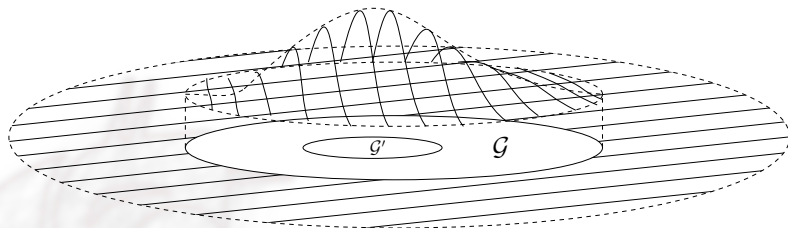


Figure: What we want a target distribution to look like in our case

Biased Proposal Step

If our goal is to bias toward similarly clustered graphs, we can utilize ARI in an energy function:

$$\begin{aligned}\alpha &= \min \left\{ 1, \frac{f(x')}{f(x_t)} \right\} \rightarrow \alpha = \min \left\{ 1, \exp \left[\frac{1}{T} (f(x') - f(x_t)) \right] \right\} \\ &\rightarrow \alpha = \min \left\{ 1, \exp \left[\frac{1}{T} (ARI(x_G, x') - ARI(x_G, x_t)) \right] \right\}\end{aligned}$$

where T is a “temperature” parameter that allows for control over movement in the chain.

Heuristically, we want T to be small enough to favor good states and big enough to explore the state space (e.g. such that acceptance rate $\approx 20\%$). In our case, $T = .01$ did the trick.

Biased Target Distribution

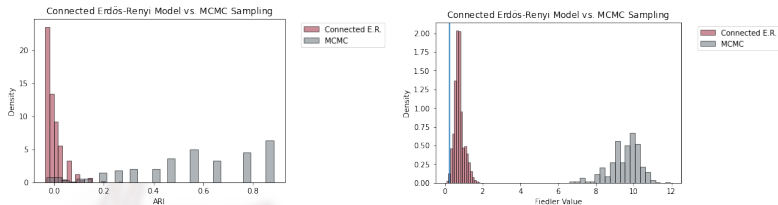


Figure: Results of 10 chains with 10,000 iterations each using energy function. $\hat{R} \approx .90$. $ESS \approx 1,200$.

Questions

Thanks!